

Incompressible Flow Solver

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1 Cartesian Mesh and Governing Equations

In this section, we will derive the discretized operators for a 2D problem flow configuration over a cartesian mesh (shown in figure 1). The equations will be spatially discretized using finite volumes on a fully staggered grid to ensure primary conservation and to avoid pressure checkerboarding.

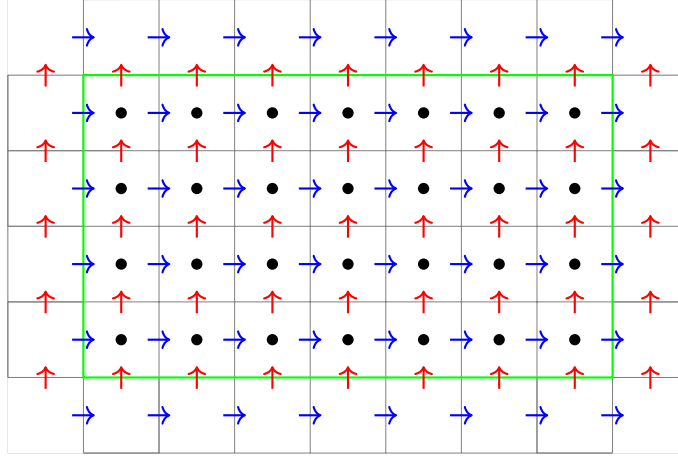


Figure 1: Staggered grid. The green line denotes the boundaries of the physical domain. Velocity and pressure values *inside* the blue domain are active degrees of freedom that must be solved for. Velocities *on* or *outside* the blue lines are boundary/ghost points that can be determined algebraically given the boundary conditions and the interior values. Blue arrows indicate x -velocities, red arrows y -velocities, and black dots denote pressure.

Before proceeding we recall the conservative form of the incompressible Navier-Stokes along with continuity.

$$\partial_t \mathbf{u} + \nabla \cdot (\mathbf{u} \mathbf{u}^\top) = -\nabla p + Re^{-1} \nabla^2 \mathbf{u} \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0 \quad (2)$$

The integral form of the equations is obtained by integrating over the volume. The divergence theorem is used to turn the volume integrals into surface integrals, where $\hat{\mathbf{n}}$ denotes the outward-pointing, unit-norm vector normal to the surface. (In our 2D setting over a cartesian grid, we have $\hat{\mathbf{n}} = (1, 0)$ or $\hat{\mathbf{n}} = (0, 1)$.)

$$\int_{\mathcal{V}} \partial_t \mathbf{u} d\mathcal{V} + \int_{\mathcal{S}} \mathbf{u} (\mathbf{u} \cdot \hat{\mathbf{n}}) d\mathcal{S} = - \int_{\mathcal{S}} p \hat{\mathbf{n}} d\mathcal{S} + Re^{-1} \int_{\mathcal{S}} \hat{\mathbf{n}} \cdot \nabla \mathbf{u} d\mathcal{S} \quad (3)$$

$$\int_{\mathcal{S}} \mathbf{u} \cdot \hat{\mathbf{n}} d\mathcal{S} = 0. \quad (4)$$

1.1 x -momentum equation in conservative form

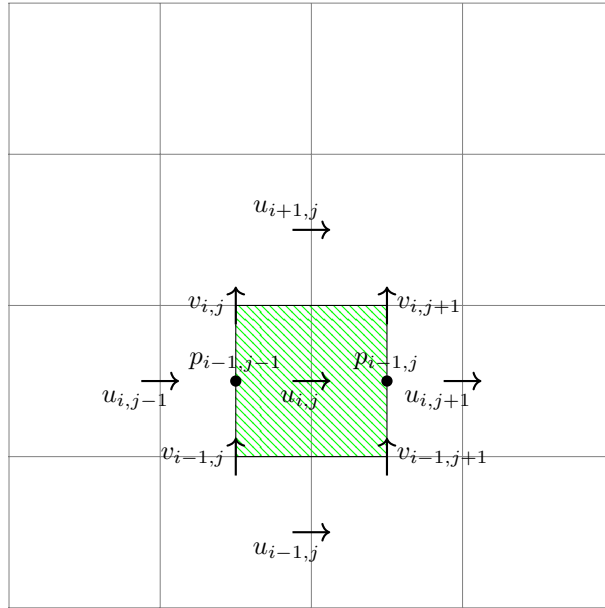


Figure 2: Control volume for x -momentum. The grid spacing in the x direction is Δx , and Δy in the y direction.

For this derivation we will refer to figure 2. Let the location of the top and bottom faces of the shaded box be y_t and y_b . Similarly, let the location of the right and left faces be x_r and x_l . Starting from equation (3) in the x direction, we have

$$\begin{aligned} \int_{\mathcal{V}} \partial_t u d\mathcal{V} + \int_{\mathcal{Y}} u^2(x_r) - u^2(x_l) d\mathcal{Y} + \int_{\mathcal{X}} (uv)(y_t) - (uv)(y_b) d\mathcal{X} = \\ - \int_{\mathcal{Y}} p(x_r) - p(x_l) d\mathcal{Y} + Re^{-1} \left(\int_{\mathcal{Y}} (\partial_x u)(x_r) - (\partial_x u)(x_l) d\mathcal{Y} + \int_{\mathcal{X}} (\partial_y u)(y_t) - (\partial_y u)(y_b) d\mathcal{X} \right). \end{aligned} \quad (5)$$

In discrete form, using the control volume in figure 2, we have

$$\begin{aligned}
\int_{\mathcal{V}} \partial_t u \, d\mathcal{V} &= \partial_t u_{i,j} \Delta x \Delta y \\
\int_{\mathcal{Y}} u^2(x_r) - u^2(x_l) \, d\mathcal{Y} &= \frac{\Delta y}{4} \left((u_{i,j+1} + u_{i,j})^2 - (u_{i,j-1} + u_{i,j})^2 \right) \\
\int_{\mathcal{X}} (uv)(y_t) - (uv)(y_b) \, d\mathcal{X} &= \frac{\Delta x}{4} \left((u_{i+1,j} + u_{i,j})(v_{i,j} + v_{i,j+1}) - (u_{i-1,j} + u_{i,j})(v_{i-1,j} + v_{i-1,j+1}) \right) \\
\int_{\mathcal{Y}} p(x_r) - p(x_l) \, d\mathcal{Y} &= (p_{i-1,j} - p_{i-1,j-1}) \Delta y \\
Re^{-1} \int_{\mathcal{Y}} (\partial_x u)(x_r) - (\partial_x u)(x_l) \, d\mathcal{Y} &= \frac{Re^{-1}}{\Delta x} (u_{i,j+1} - 2u_{i,j} + u_{i,j-1}) \Delta y \\
Re^{-1} \int_{\mathcal{X}} (\partial_y u)(y_t) - (\partial_y u)(y_b) \, d\mathcal{X} &= \frac{Re^{-1}}{\Delta y} (u_{i+1,j} - 2u_{i,j} + u_{i-1,j}) \Delta x
\end{aligned} \tag{6}$$

Putting this all together and dividing through by $\Delta x \Delta y$, we obtain

$$\begin{aligned}
&\partial_t u_{i,j} + \frac{1}{4\Delta x} \left((u_{i,j+1} + u_{i,j})^2 - (u_{i,j-1} + u_{i,j})^2 \right) \\
&+ \frac{1}{4\Delta y} \left((u_{i+1,j} + u_{i,j})(v_{i,j} + v_{i,j+1}) - (u_{i-1,j} + u_{i,j})(v_{i-1,j} + v_{i-1,j+1}) \right) \\
&= -\frac{p_{i-1,j} - p_{i-1,j-1}}{\Delta x} + Re^{-1} \left(\frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta x^2} + \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta y^2} \right).
\end{aligned}$$

1.2 y -momentum equation in conservative form

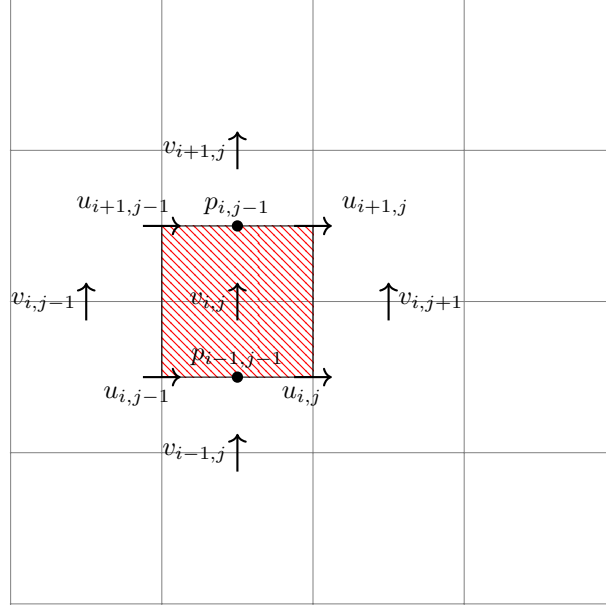


Figure 3: Control volume for y -momentum

We may recognize that the y -momentum is functionally symmetric to the x -momentum. It therefore follows that Equation (3) allows us to obtain

$$\int_{\mathcal{V}} \partial_t v \, d\mathcal{V} + \int_{\mathcal{X}} \left(v^2(y_t) - v^2(y_b) \right) d\mathcal{X} + \int_{\mathcal{Y}} \left((uv)(x_r) - (uv)(x_l) \right) d\mathcal{Y} \quad (7)$$

$$= - \int_{\mathcal{X}} \left(p(y_t) - p(y_b) \right) d\mathcal{X} + Re^{-1} \left(\int_{\mathcal{X}} \left((\partial_y v)(y_t) - (\partial_y v)(y_b) \right) d\mathcal{X} + \int_{\mathcal{Y}} \left((\partial_x v)(x_r) - (\partial_x v)(x_l) \right) d\mathcal{Y} \right). \quad (8)$$

Using the same discretization steps, we may integrate and divide by $\Delta x \Delta y$ to obtain,

$$\begin{aligned} & \partial_t v_{i,j} + \frac{1}{4\Delta y} \left((v_{i+1,j} + v_{i,j})^2 - (v_{i-1,j} + v_{i,j})^2 \right) \\ & + \frac{1}{4\Delta x} \left((u_{i,j+1} + u_{i,j})(v_{i,j+1} + v_{i,j}) - (u_{i,j-1} + u_{i,j})(v_{i,j-1} + v_{i,j}) \right) \\ & = - \frac{p_{i+1,j} - p_{i-1,j}}{\Delta y} + Re^{-1} \left(\frac{v_{i,j+1} - 2v_{i,j} + v_{i,j-1}}{\Delta x^2} + \frac{v_{i+1,j} - 2v_{i,j} + v_{i-1,j}}{\Delta y^2} \right). \end{aligned}$$

1.3 Continuity equation

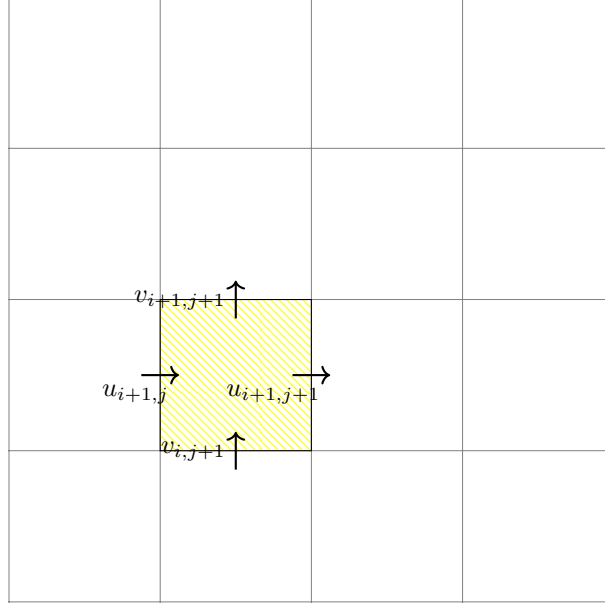


Figure 4: Control volume for the continuity equation.

Starting from equation (4), and using the control volume in figure 4, we have

$$(u_{i+1,j+1} - u_{i+1,j}) \Delta y + (v_{i+1,j+1} - v_{i,j+1}) \Delta x = 0. \quad (9)$$

Dividing through by Δx and Δy , we obtain

$$\boxed{\frac{1}{\Delta x} (u_{i+1,j+1} - u_{i+1,j}) + \frac{1}{\Delta y} (v_{i+1,j+1} - v_{i,j+1}) = 0.}$$